

Introduction to TACC Computing Cluster

An abstract digital artwork featuring a dark blue background with glowing purple and orange circuit lines. A large, stylized knight chess piece, rendered in orange and blue, is positioned on the right side, appearing to emerge from or interact with the circuitry. The overall aesthetic is futuristic and technological.

Scalable Computational Intelligence
Texas Advanced Computing Center (TACC)

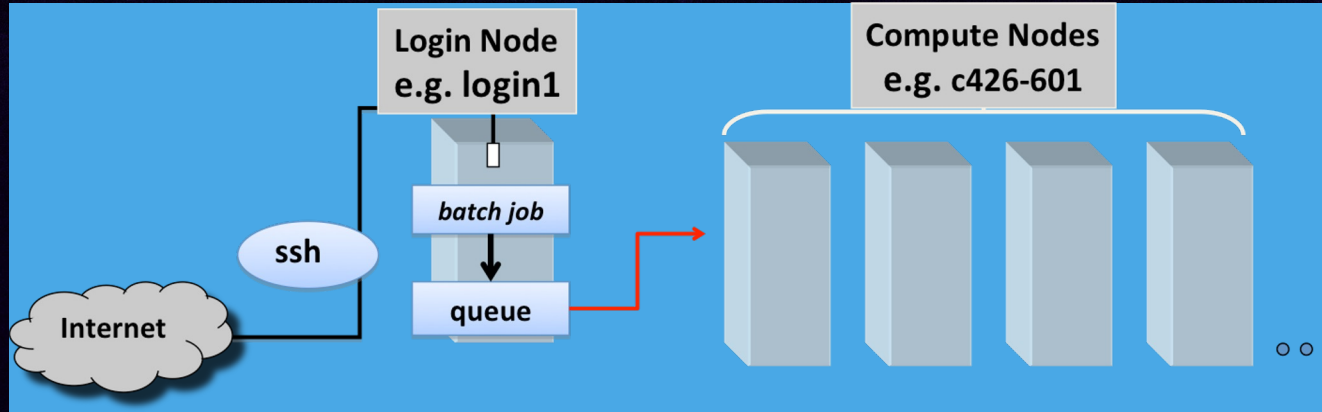
Overview

Practical introduction to using TACC resources

- Nodes and File system at TACC
- How to Remotely Access TACC
- Navigating the File System on TACC machines
- Hands On: Accessing TACC via TAP



Login vs Compute Nodes



Login Nodes

- Shared by multiple users
- Not meant for CPU or memory intensive jobs
- Meant for managing file and compiling code

Compute Nodes

- Where research computation occurs
- Training machine learning models, running simulations, etc.

Types of Compute Nodes: Partitions

- Compute nodes are divided into different pools called queues
 - For example here are a few queues on Frontera
 - development (interactive sessions)
 - small (one or two node jobs)
 - normal (≥ 3 node jobs)
 - rtx (gpu)
 - rtx-dev (interactive sessions with gpus)
 - nvdimm (large memory nodes)
 - Refer to the user guide for the queue and accounting details



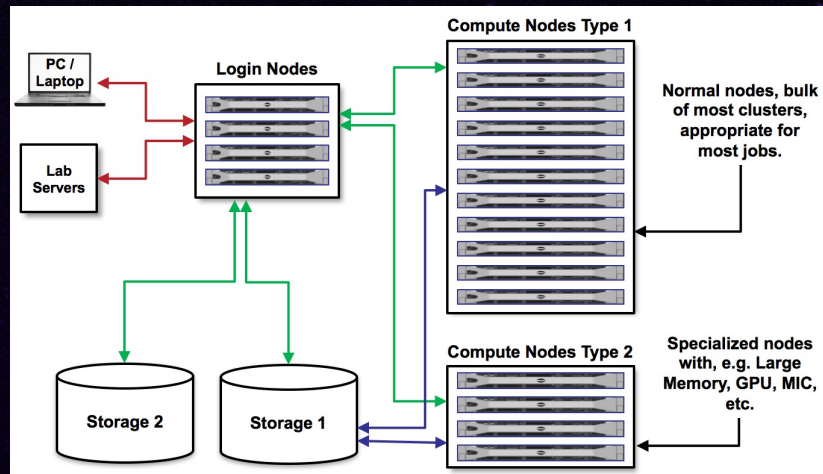
File Systems

Three (3) main file systems

- **\$HOME**
 - Quota (25G)
 - Backed up
 - Used for: small scripts, environment settings, and other routine file management task
- **\$WORK**
 - Quota (1TB)
 - Not Backed up
 - Used For: Intermediate staging/preparation of files, storing original datasets
- **\$SCRATCH**
 - No Quota
 - Not Backed up, subject to purge
 - Used for: High I/O bandwidth jobs

Refer to Frontera User Guide / TACC tutorials

- <https://frontera-portal.tacc.utexas.edu/user-guide/files/>
- <https://docs.tacc.utexas.edu/tutorials/managingio/>

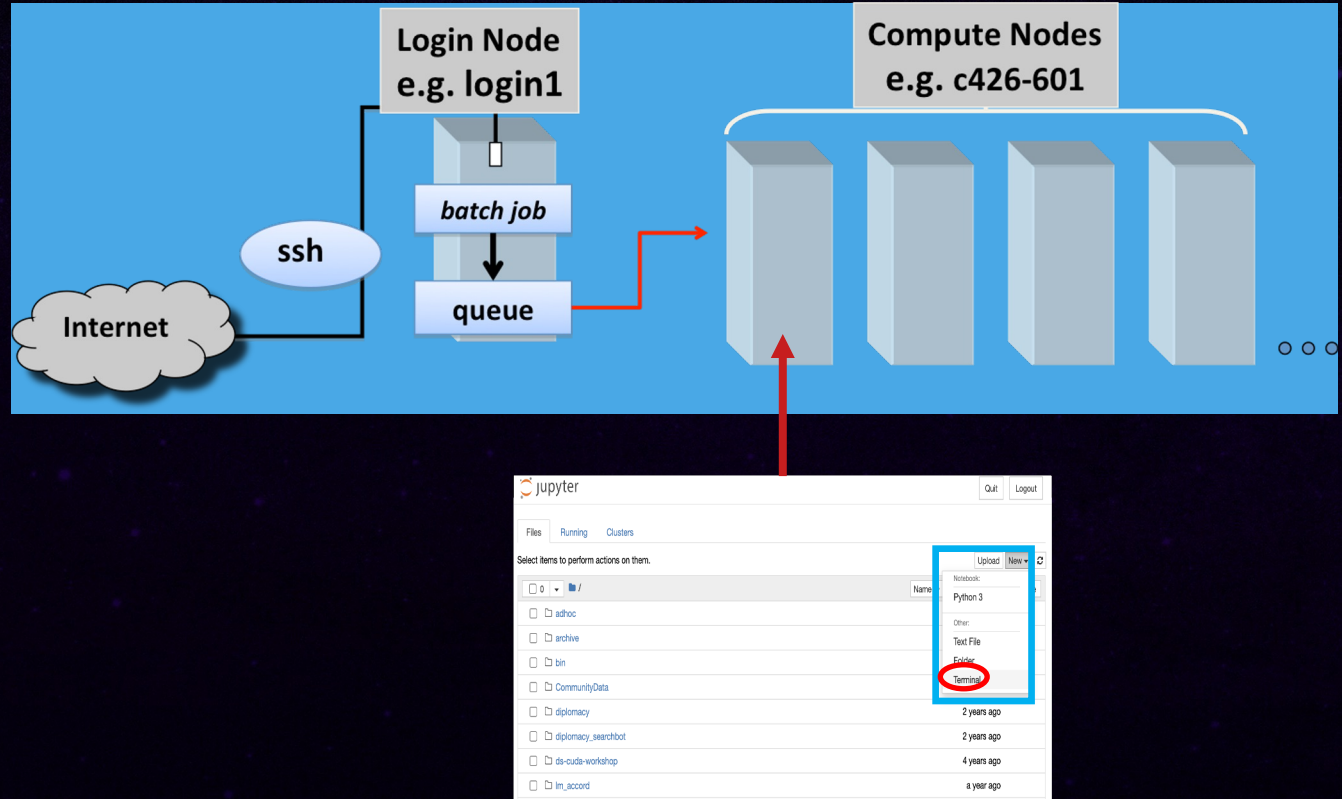


- **/tmp**
 - Each compute node comes with local storage (/tmp)
 - Fast I/O
 - **Temporary:** storage only last the duration of the job and should be copied for permanent storage

How to Remotely Access TACC

We will remotely access Frontera in two ways in this event:

1. SSH: Access login nodes via command line
2. TACC Analysis Portal: UI for accessing jupyter notebook session on a compute nodes



Accessing Compute Nodes

Let's assume we have logged into TACC via ssh and we are on a log in node. How do we connect to a compute node?

- submit a batch job via the sbatch command
 - job waits in queue until resources are available
- activate an interactive session via **idev**
 - idev
 - Allows you to get a compute node for testing purposes
 - -m option lets you specify time in minutes
 - idev -m 30



Bash – basic file commands

- TACC machines use the Linux operating system
- Default shell and most popular is bash (Bourne Again Shell)

Command	Function
<code>pwd</code>	Shows absolute path of your present working directory
<code>ls <directory name></code>	Lists a directory (if mentioned) or present working directory
<code>mkdir <name></code>	Creates a new directory with mentioned name
<code>cd <name></code>	Changes working directory to name mentioned (or go to home directory if nothing is specified)
<code>cp <File1> <File2></code> <code>cp <File1> <Directory></code>	Makes a copy of File1 and names it File2 or makes a copy of the file with the same name in Directory1
<code>cp -r <directory1> <directory2></code>	Makes a copy of directory1 and all its contents into directory2
<code>mv <File1> <Directory></code> <code>mv <File1> <File2></code>	moves File1 to Directory1 or rename File1 to File2
<code>rm <File1></code> <code>rm -r <Directory1></code>	Removes a file or Remove a directory (add <code>-f</code> flag to force remove...be careful)

module

- On TACC Systems the module system helps setup or teardown useful packages that are supported by TACC, on the fly
- To find more information on a module
 - `module spider package-name`
- To load a module
 - `module load package-name`
 - `module load package-name/version.number`

Check out the User guide to learn more:

https://lmod.readthedocs.io/en/latest/010_user.html



Hands on: Accessing TACC via the TACC Analysis Portal and Environment Setup

Development Environment

Jupyter Notebook

<http://jupyter.org/>



Web application for creating and sharing computational documents.

It offers a simple, streamlined, document-centric experience.

Provides an interactive interface in the browser

- Can view output and plots inline

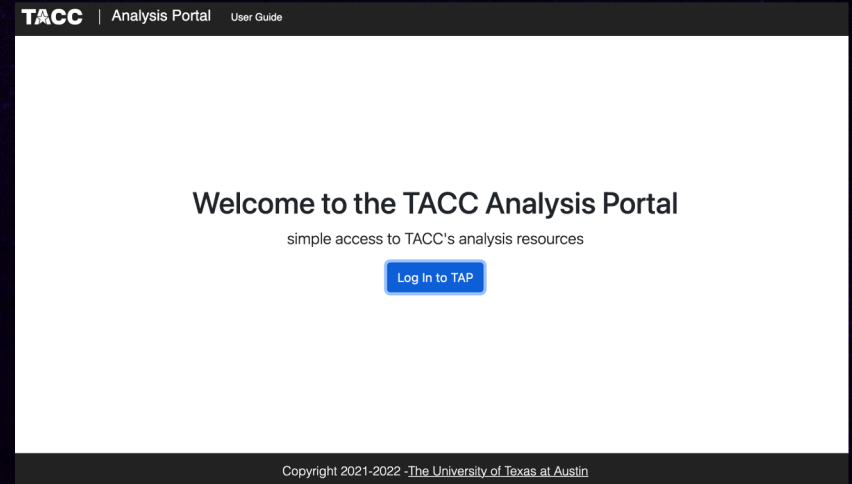
- Makes Iterative/Interactive Development easy

- Can be turned into a pure script for non-interactive work



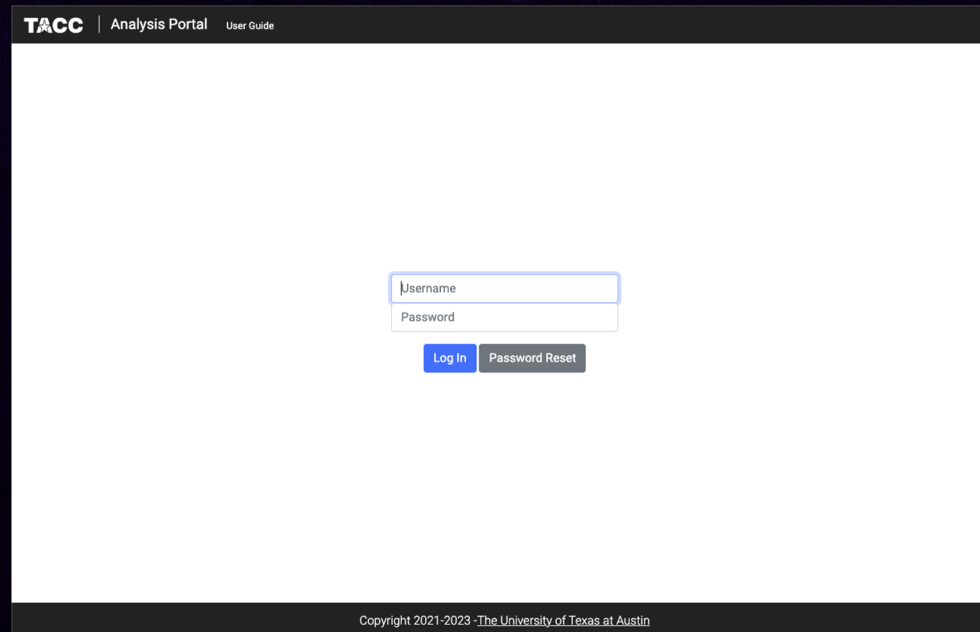
TACC Analysis Portal

- TACC Analysis Portal (TAP) provides access to interactive sessions on TACC machines
- Allows user to utilize web-based interactions on TACC computers
 - e.g. Jupyter notebook, Rstudio, remote linux desktop
- <https://tap.tacc.utexas.edu>



Login with your TACC account

- TACC Analysis Portal:
login with your TACC account credentials



The screenshot shows the TACC Analysis Portal login interface. At the top, a dark header bar contains the TACC logo, 'Analysis Portal', and a 'User Guide' link. The main content area is white and features a login form with two input fields: 'Username' and 'Password'. Below these fields are two buttons: a blue 'Log In' button and a grey 'Password Reset' button. At the bottom of the page, a dark footer bar displays the copyright text: 'Copyright 2021-2023 -The University of Texas at Austin'.

Jupyter notebook: Accessing shell

TACC | Analysis Portal User Guide jrduncan Log Out

Submit New Job

System

Application

Select System

Project

Select System

Queue

Select System

Nodes

1

Tasks

1

Options

Job Name

20 characters max

Time Limit

H:M:S (default 2:0:0)

Reservation

reservation name

VNC Desktop Resolution

WIDTHxHEIGHT

Submit

Utilities

System Status

System	Status	Utilization	Job Count
Frontera	✓ Open	99%	Running: 312 Queued: 1251
Lonestar6	✓ Open	69%	Running: 135 Queued: 98
Longhorn	✓ Open	74%	Running: 30 Queued: 40
Maverick2	✓ Open	16%	Running: 4 Queued: 7
Stampede2	✓ Open	96%	Running: 830 Queued: 736

Past Jobs

JNB-Frontera	03/18/2022	Details	Resubmit
JNB-Frontera	03/18/2022	Details	Resubmit
JNB-Frontera	03/18/2022	Details	Resubmit
JNB-Frontera	03/18/2022	Details	Resubmit
JNB-Frontera	02/21/2022	Details	Resubmit

TACC
TEXAS ADVANCED COMPUTING CENTER


The University of Texas at Austin

Starting Jupyter notebook: Accessing shell

TACC | Analysis Portal [User Guide](#) gj3385 [Log Out](#)

Submit New Job

System

Vista

Application

Jupyter notebook

Project

TRA24005

Queue

gh

Nodes

1

Tasks

1

Options

Job Name

20 characters max

Time Limit

H:M:S (default 2:0:0)

Reservation

PEARC26-AI-AGENTS

VNC Desktop Resolution

WIDTHxHEIGHT

[Submit](#)

 **CLICK!**

System Status

System	Status	Utilization	Job Count
Frontera Details	✓ Open	91%	Running: 440 Waiting: 32
Lonestar6 Details	✓ Open	91%	Running: 326 Waiting: 1118
Stampede3 Details	✓ Open	99%	Running: 469 Waiting: 84
Vista Details	✓ Open	100%	Running: 273 Waiting: 9

Past Jobs

JNB-Vista	07/20/2026	Details	Resubmit
JNB-Vista	07/20/2026	Details	Resubmit
JNB-Vista	07/17/2026	Details	Resubmit
JNB-Vista	07/16/2026	Details	Resubmit

Waiting for Job to start (can take a minute)

TACC | Analysis Portal User Guide ngaffney

TAP Job Status

Job: Jupyter notebook on Frontera (6314485, 2024-05-13T13:42:05:00)

Status: PENDING

Refresh: in 41 seconds

Message:

TAP: Your script has been submitted to Frontera but it is not yet running.

``sinfo -p development` output is:`

PARTITION	AVAIL NODES(A/I)
development*	up 390/4

Check Status

End Job

Show Output

Back to Jobs

Jupyter notebook: Accessing shell

TACC | Analysis Portal User Guide jrduncan Log Out

TAP Job Status

Job: Jupyter notebook on Frontera (4175197, 2022-03-21T17:28-05:00)
Status: RUNNING
Start: March 21, 2022, 5:28 p.m.
End: March 21, 2022, 5:33 p.m.
Refresh: in 873 seconds
Message:
TAP: Your session is running at <https://frontera.tacc.utexas.edu:60752/?token=9cbad0f26752e7dd14fcf090d6a30b6ec5c15c63ed7d9e2b626f214712fb8b4d>

Connect End Job Show Output Back to Jobs

Setting up for the tutorial

Note, the following setup only has to be done once.
This is copying files into the right places so you can run these notebooks with us.

Jupyter notebook: Accessing shell



The screenshot displays the JupyterLab web interface. At the top, the 'jupyter' logo is on the left, and 'Quit' and 'Logout' buttons are on the right. Below the logo, there are tabs for 'Files', 'Running', and 'Clusters'. The 'Files' tab is active, showing a list of files and folders. A message 'Select items to perform actions on them.' is displayed above the file list. The file list includes folders like 'adhoc', 'archive', 'bin', 'CommunityData', 'diplomacy', 'diplomacy_searchbot', 'ds-cuda-workshop', and 'lm_accord'. A 'New' button is visible in the top right of the file list area. A dropdown menu is open from the 'New' button, showing options: 'Notebook: Python 3', 'Other: Text File', 'Folder', and 'Terminal'. The 'Terminal' option is highlighted with a red circle. The 'Upload' and 'New' buttons are also visible in the top right of the file list area.

jupyter

Quit Logout

Files Running Clusters

Select items to perform actions on them.

0 /

adhoc

archive

bin

CommunityData

diplomacy

diplomacy_searchbot

ds-cuda-workshop

lm_accord

2 years ago

2 years ago

4 years ago

a year ago

Upload New

Notebook:

Python 3

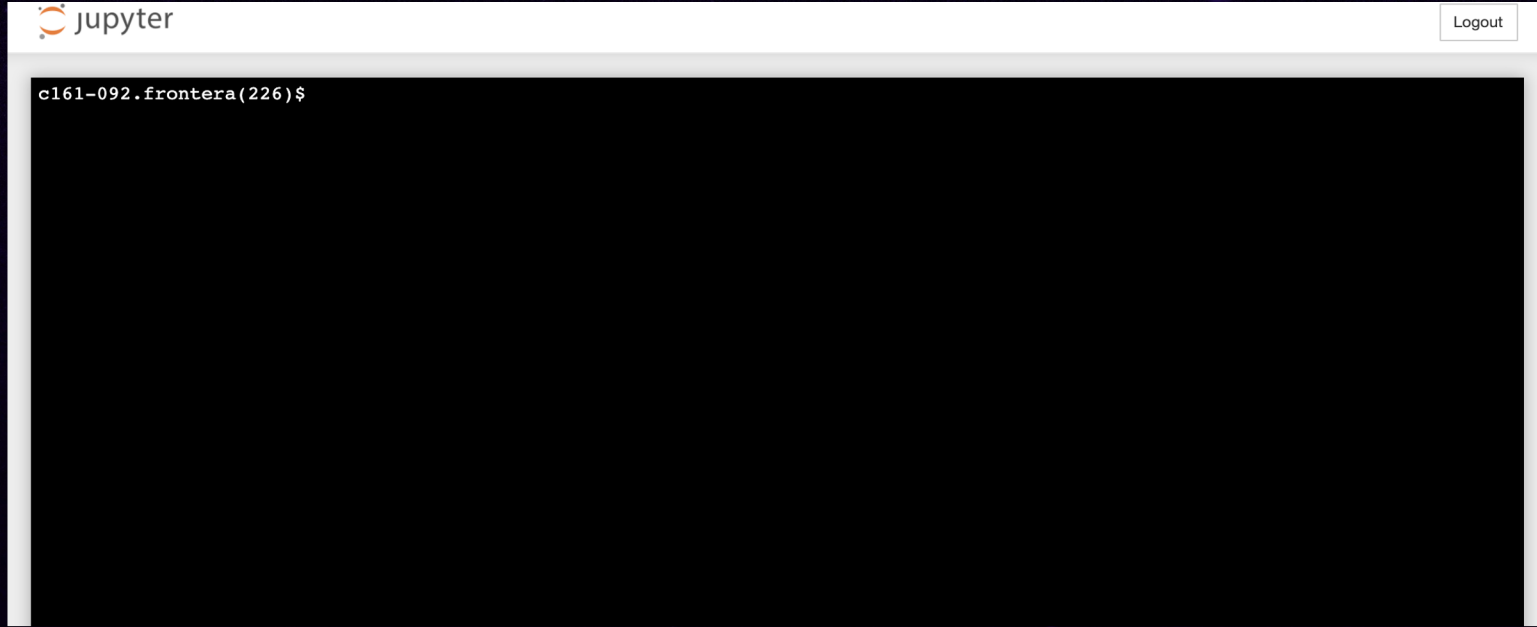
Other:

Text File

Folder

Terminal

Jupyter notebook: Accessing shell (4)



The image shows a Jupyter notebook interface. At the top left is the Jupyter logo and the word "jupyter". At the top right is a "Logout" button. The main area is a black terminal window with the text "c161-092.frontera(226)\$" in white, indicating a shell prompt.

```
jupyter
```

Logout

```
c161-092.frontera(226)$
```



One time installation

From the command line enter

```
/scratch1/10386/lsmith9003/share/container_setup_ml_institute_summer_2025/install
```

After running this enter **ls** and make sure you see an entry for **PEARC2026**

Stop the Session

TACC | Analysis Portal | User Guide | **ngaffney** | Log Out

TAP Job Status

Job: Jupyter notebook on Frontera (6301729, 2024-05-07T11:49-05:00)
Status: RUNNING
Start: 05/07/2024 11:49
End: 05/07/2024 13:49
Refresh: in 630 seconds

Message:

TAP: Your session is running at <https://frontera.tacc.utexas.edu:60069/?token=H7Y2IZVbCVPpJRn9-U9tArNnvynEX3D0tP29pqnmoYc>

Connect

End Job

Show Output

Back to Jobs

Back to the Jobs page

TACC | Analysis Portal | User Guide | **ngaffney** | Log Out

TAP Job Status

Job: Jupyter notebook on Frontera (6301729, 2024-05-07T11:49-05:00)
Status: RUNNING
Start: 05/07/2024 11:49
End: 05/07/2024 13:49
Refresh: in 630 seconds

Message:

TAP: Your session is running at <https://frontera.tacc.utexas.edu:60069/?token=H7Y2IZVbCVPpJRn9-U9tArNnvynEX3D0tP29pqnMOYc>

Connect

End Job

Show Output

Back to Jobs

This is the end of the one time setup. Next we will make sure it worked and then get started with some notebooks about Python

Jupyter notebook: Accessing shell

TACC Visualization Portal:

Go to <https://tap.tacc.utexas.edu>; Login with your training account
credentials

TACC | Analysis Portal | User Guide

jrduncan

Log Out

Submit New Job

System

Application

Select System

Project

Select System

Queue

Select System

Nodes

1

Tasks

1

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JNB-Frontera	02/21/2022	Details	Resubmit

Jupyter notebook: Accessing shell

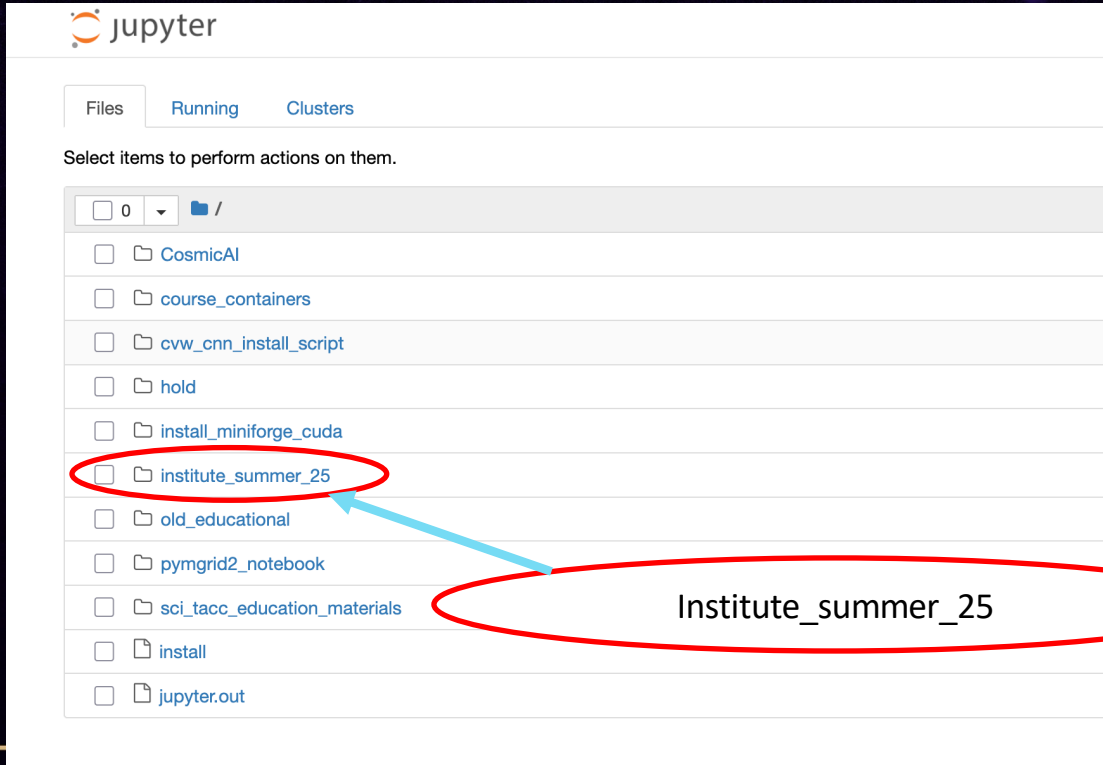
TACC | Analysis Portal User Guide jrduncan Log Out

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Connect End Job Show Output Back to Jobs

Jupyter notebook: Accessing New Python Notebook



The screenshot shows the JupyterLab interface with the 'Files' tab selected. The 'Running' and 'Clusters' tabs are also visible. Below the tabs, there is a prompt: 'Select items to perform actions on them.' The file browser displays a list of folders and files. The folder 'institute_summer_25' is highlighted with a red circle. A blue arrow points from this folder to a larger red circle containing the text 'Institute_summer_25'.

jupyter

Files Running Clusters

Select items to perform actions on them.

0 /

- ☐ CosmicAI
- ☐ course_containers
- ☐ cvw_cnn_install_script
- ☐ hold
- ☐ install_miniforge_cuda
- ☐ **institute_summer_25**
- ☐ old_educational
- ☐ pymgrid2_notebook
- ☐ sci_tacc_education_materials
- ☐ install
- ☐ jupyter.out

Institute_summer_25

Jupyter Notebook: Open Existing File

- Navigate to
[PEARC2026/pearc26_tutorial_ai_agents/03_intro_building_ai_agents](#)
- Click on the file notebook file [Introduction to AI Agents.ipynb](#)
- A Jupyter notebook should launch in a new tab
- Check to make sure you can run this notebook in the containerized kernel



Thanks!!

- Any Issues on TACC Systems that we can not help you with?
 - Open a ticket at <https://tacc.utexas.edu/about/help/>



Supplemental Slides

File Transfer Mechanisms

- Copying files from your local machine to TACC system
- Mac/OSX or Linux
 - Open Terminal window
 - Use scp (Secure Shell Copy)
 - Copies over ssh connection
 - -r option for recursive



scp [local file name] user@frontera.tacc.utexas.edu

scp -r [local folder name] user@frontera.tacc.utexas.edu

File Transfer Mechanisms (2)

- Copying files from your local machine to TACC system
- Windows
 - Download WinSCP (<https://winscp.net>)
 - Double click installer
 - Choose “Explorer” Option
 - Hostname: `frontera.tacc.utexas.edu`
 - Username: `username`
 - TACC Code: <Use Code from your credentials handout>
- Drag and Drop Files/Folders to copy to Stampede2



Command Line Editing

- Often will be required
 - Working with code
 - Editing configuration files
 - Writing Bash Scripts
 - TACC Systems: Writing job submission scripts
- Vim
 - Vi Improved
- Emacs
- Nano
 - Simple, Easy to use



module

- On TACC Systems the module system helps setup or teardown useful packages that are supported by TACC, on the fly
- To find more information on a module
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 - `module load package-name`
 - `module load package-name/version.number`
- [Learn more here](#)

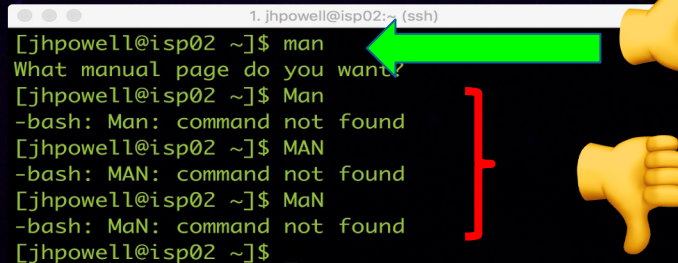


Linux/bash

- TACC machines use the Linux operating system
- Default shell and most popular is bash (Bourne Again Shell)
- Bash features auto completion of commands with tab key
- Bash commands are case sensitive

Example:
The “**man**”
command

`man` $\neq$ `Man` $\neq$ `MAN`



A terminal window titled '1. jhpowell@isp02:~ (ssh)' showing the following commands and outputs:

```
[jhpowell@isp02 ~]$ man  
What manual page do you want?  
[jhpowell@isp02 ~]$ Man  
-bash: Man: command not found  
[jhpowell@isp02 ~]$ MAN  
-bash: MAN: command not found  
[jhpowell@isp02 ~]$ MaN  
-bash: MaN: command not found  
[jhpowell@isp02 ~]$ _
```

A green arrow points to the first 'man' command, and a red bracket groups the subsequent three failed attempts. A thumbs up emoji is next to the first line, and a thumbs down emoji is next to the failed attempts.